

Questions with Answer Keys

MathonGo

Q1 - 2024 (01 Feb Shift 1)

The area enclosed by the curves $xy + 4y = 16$ and $x + y = 6$ is equal to :

(1) $28 - 30 \log_e 2$

(2) $30 - 28 \log_e 2$

(3) $30 - 32 \log_e 2$

(4) $32 - 30 \log_e 2$

Q2 - 2024 (01 Feb Shift 2)

Three points $O(0, 0)$, $P(a, a^2)$, $Q(-b, b^2)$, $a > 0$, $b > 0$, are on the parabola $y = x^2$. Let S_1 be the area of the region bounded by the line PQ and the parabola, and S_2 be the area of the triangle OPQ . If the minimum value of $\frac{S_1}{S_2}$ is $\frac{m}{n}$, $\gcd(m, n) = 1$, then $m + n$ is equal to :

Q3 - 2024 (01 Feb Shift 2)

The sum of squares of all possible values of k , for which area of the region bounded by the parabolas

$2y^2 = kx$ and $ky^2 = 2(y - x)$ is maximum, is equal to :

Q4 - 2024 (27 Jan Shift 1)

Let the area of the region $\{(x, y) : x - 2y + 4 \geq 0, x + 2y^2 \geq 0, x + 4y^2 \leq 8, y \geq 0\}$ be $\frac{m}{n}$, where m and n are coprime numbers. Then $m + n$ is equal to _____

Q5 - 2024 (27 Jan Shift 2)

If the area of the region $\{(x, y) : 0 \leq y \leq \min\{2x, 6x - x^2\}\}$ is A , then $12A$ is equal to.

Q6 - 2024 (29 Jan Shift 1)

The area (in sq. units) of the part of circle $x^2 + y^2 = 169$ which is below the line $5x - y = 13$ is

$\frac{\pi\alpha}{2\beta} - \frac{65}{2} + \frac{\alpha}{\beta} \sin^{-1}\left(\frac{12}{13}\right)$ where α, β are coprime numbers. Then $\alpha + \beta$ is equal to

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Q7 - 2024 (29 Jan Shift 2)

Let the area of the region $\{(x, y) : 0 \leq x \leq 3, 0 \leq y \leq \min\{x^2 + 2, 2x + 2\}\}$ be A . Then $12A$ is equal to

Q8 - 2024 (30 Jan Shift 1)

The area (in square units) of the region bounded by the parabola $y^2 = 4(x - 2)$ and the line $y = 2x - 8$

(1) 8

(2) 9

(3) 6

(4) 7

Q9 - 2024 (30 Jan Shift 2)

Let $Y = Y(X)$ be a curve lying in the first quadrant such that the area enclosed by the line $Y - y = Y'(x)(X - x)$ and the co-ordinate axes, where (x, y) is any point on the curve, is always

$\frac{-y^2}{2Y'(x)} + 1, Y'(x) \neq 0$. If $Y(1) = 1$, then $12Y(2)$ equals _____.

Q10 - 2024 (30 Jan Shift 2)

The area of the region enclosed by the parabola $(y - 2)^2 = x - 1$, the line $x - 2y + 4 = 0$ and the positive coordinate axes is _____.

Q11 - 2024 (31 Jan Shift 1)

The area of the region $\{(x, y) : y^2 \leq 4x, x < 4, \frac{xy(x-1)(x-2)}{(x-3)(x-4)} > 0, x \neq 3\}$ is

(1) $\frac{16}{3}$ (2) $\frac{64}{3}$ (3) $\frac{8}{3}$ (4) $\frac{32}{3}$

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Questions with Answer Keys

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Q12 - 2024 (31 Jan Shift 2)

The area of the region enclosed by the parabola $y = 4x - x^2$ and $3y = (x - 4)^2$ is equal to

(1) $\frac{32}{9}$

(2) 4

(3) 6

(4) $\frac{14}{3}$

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Answer Key

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Solutions

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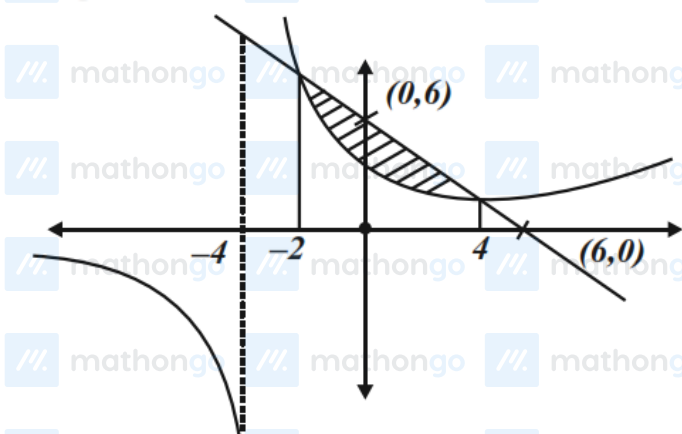
Q1

$$xy + 4y = 16, \quad x + y = 6$$

$$y(x + 4) = 16$$

$$x + y = 6$$

on solving, (1) & (2)

we get $x = 4, x = -2$ 

$$\begin{aligned} \text{Area} &= \int_{-2}^4 \left((6 - x) - \left(\frac{16}{x+4} \right) \right) dx \\ &= 30 - 32 \ln 2 \end{aligned}$$

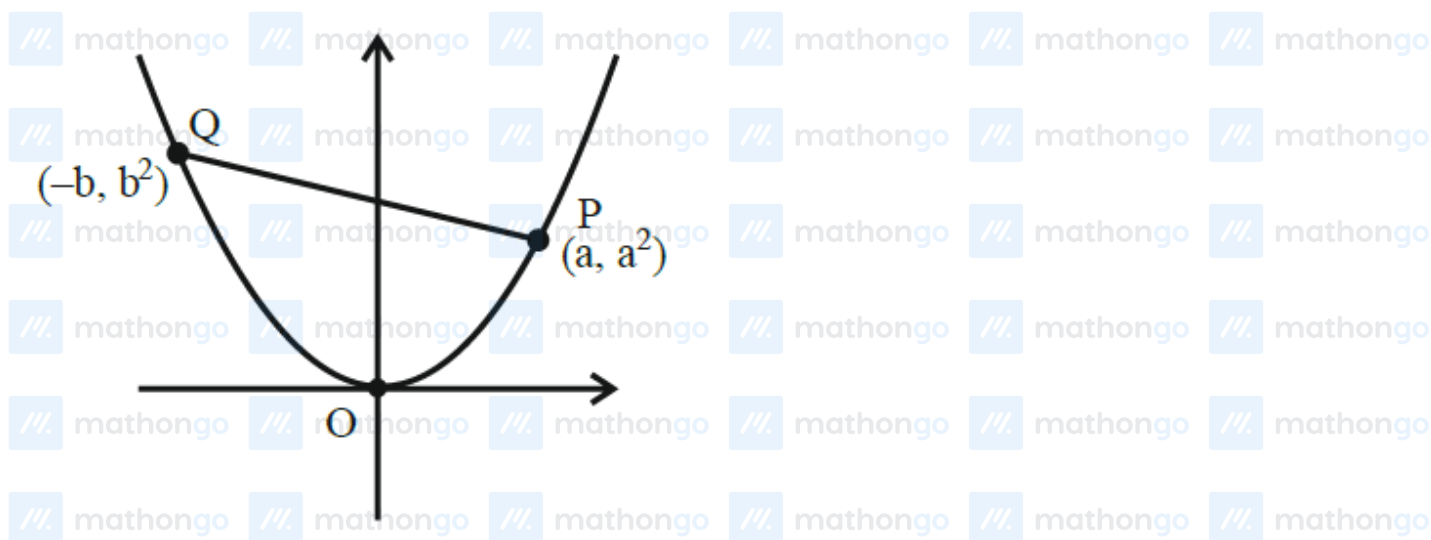
Q2

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Solutions

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$$S_2 = \frac{1}{2} \begin{vmatrix} 0 & 0 & 1 \\ a & a^2 & 1 \\ -b & b^2 & 1 \end{vmatrix} = \frac{1}{2} (ab^2 + a^2b)$$

$$PQ: -y - a^2 = \frac{a^2 - b^2}{a + b}(x - a)$$

$$y - a^2 = (a - b)x - (a - b)a$$

$$y = (a - b)x + ab$$

$$S_1 = \int_{-b}^a ((a - b)x + ab - x^2) dx$$

$$= (a - b) \frac{x^2}{2} + (ab)x - \frac{x^3}{3} \Big|_{-b}^a$$

$$= \frac{(a - b)^2(a + b)}{2} + ab(a + b) - \frac{(a^3 + b^3)}{3}$$

$$\frac{S_1}{S_2} = \frac{\frac{(a - b)^2}{2} + ab - \frac{(a^2 + b^2 - ab)}{3}}{\frac{ab}{2}}$$

$$= \frac{3(a - b)^2 + 6ab - 2(a^2 + b^2 - ab)}{3ab}$$

$$= \frac{1}{3} \left[\frac{a}{b} + \frac{b}{a} + 2 \right]$$

$$= \frac{4}{3} = \frac{m}{n} \quad m + n = 7$$

Q3

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Solutions

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$$ky^2 = 2(y - x)$$

$$2y^2 = kx$$

Point of intersection $\rightarrow$

$$ky^2 = \left(\frac{y-2y^2}{k}\right)$$

$$y = 0 \quad ky = 2\left(\frac{1-2y}{k}\right)$$

$$ky + \frac{4y}{k} = 2$$

$$y = \frac{2}{k + \frac{4}{k}} = \frac{2k}{k^2 + 4}$$

$$A = \int_0^{\frac{2k}{k^2+4}} \left(\left(y - \frac{ky^2}{2} \right) - \left(\frac{2y^2}{k} \right) \right) \cdot dy$$

$$= \frac{y^2}{2} - \left(\frac{k}{2} + \frac{2}{k} \right) \cdot \frac{y^3}{3} \Big|_0^{\frac{2k}{k^2+4}}$$

$$= \left(\frac{2k}{k^2+4} \right)^2 \left[\frac{1}{2} - \frac{k^2+4}{2k} \times \frac{1}{3} \times \frac{2k}{k^2+4} \right]$$

$$= \frac{1}{6} \times 4 \times \left(\frac{1}{k + \frac{4}{k}} \right)^2$$

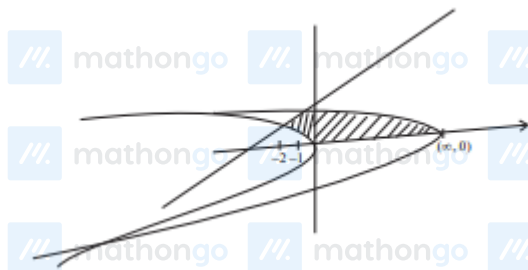
$$A \cdot M \geq G \cdot M \frac{\left(k + \frac{4}{k} \right)}{2} \geq 2$$

$$k + \frac{4}{k} \geq 4$$

Area is maximum when $k = \frac{4}{k}$

$$k = 2, -2$$

Q4



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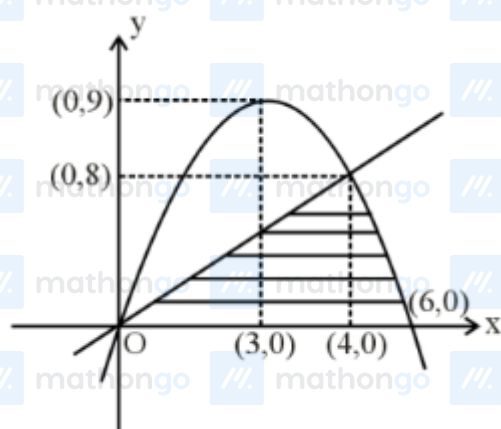
Solutions

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$$\begin{aligned}
 A &= \int_0^1 [(8 - 4y^2) - (-2y^2)] dy + \int_1^{3/2} [(8 - 4y^2) - (2y - 4)] dy \\
 &= \left[8y - \frac{2y^3}{3} \right]_0^1 + \left[12y - y^2 - \frac{4y^3}{3} \right]_1^{3/2} = \frac{107}{12} = \frac{m}{n} \\
 \therefore m + n &= 119
 \end{aligned}$$

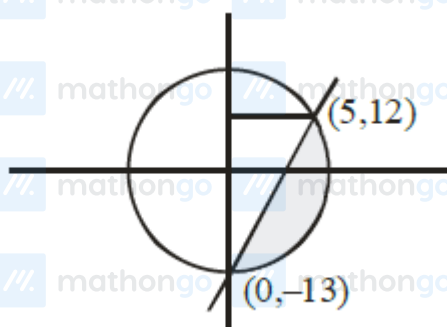
Q5

We have



$$\begin{aligned}
 A &= \frac{1}{2} \times 4 \times 8 + \int_4^6 (6x - x^2) dx \\
 A &= \frac{76}{3} \\
 12A &= 304
 \end{aligned}$$

Q6



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Solutions

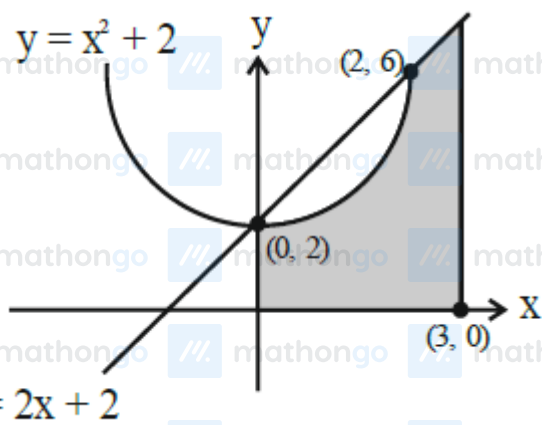
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$$\text{Area} = \int_{-13}^{12} \sqrt{169 - y^2} dy - \frac{1}{2} \times 25 \times 5$$

$$= \frac{\pi}{2} \times \frac{169}{2} - \frac{65}{2} + \frac{169}{2} \sin^{-1} \frac{12}{13}$$

$$\therefore \alpha + \beta = 171$$

Q7



$$A = \int_0^2 (x^2 + 2) dx + \int_2^3 (2x + 2) dx$$

$$A = \frac{41}{3}$$

$$12A = 41 \times 4 = 164$$

Q8

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Solutions

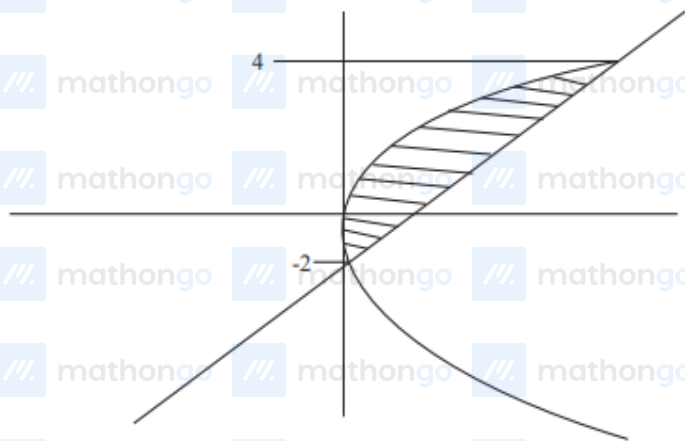
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Let $X = x - 2$

$$y^2 = 4x, \quad y = 2(x + 2) - 8$$

$$y^2 = 4x, \quad y = 2x - 4$$

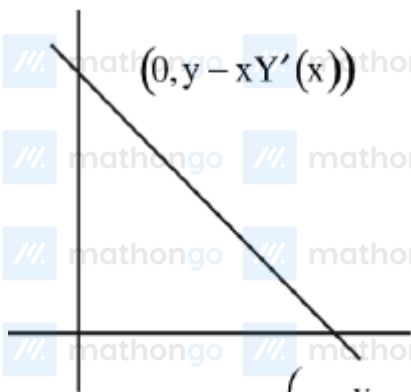
$$A = \int_{-2}^4 \frac{y^2}{4} - \frac{y + 4}{2}$$



$$= 9$$

Q9

$$A = \frac{1}{2} \left(\frac{-y}{Y'(x)} + x \right) (y - xY'/x) = \frac{-y^2}{2Y'(x)} + 1$$



$$\left(\frac{-y}{Y'(x)} + x, 0 \right)$$

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Solutions

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$$(-y + xY'(x))(y - xY'(x)) = -y^2 + 2Y'(x)$$

$$-y^2 + xyY'(x) + xyY'(x) - x^2[Y'(x)]^2$$

$$2xy - y^2 y'(x) = 2Y'(x)$$

$$\frac{dy}{dx} = \frac{2xy - 2}{x^2}$$

$$\frac{dy}{dx} - \frac{2}{x}y = \frac{-2}{x^2}$$

$$\text{I.F.} = e^{-2\ln x} = \frac{1}{x^2}$$

$$y \cdot \frac{1}{x^2} = \frac{2}{3}x^{-3} + c$$

$$\text{Put } x = 1, y = 1$$

$$1 = \frac{2}{3} + c \Rightarrow c = \frac{1}{3}$$

$$Y = \frac{2}{3} \cdot \frac{1}{x} + \frac{1}{3}X^2$$

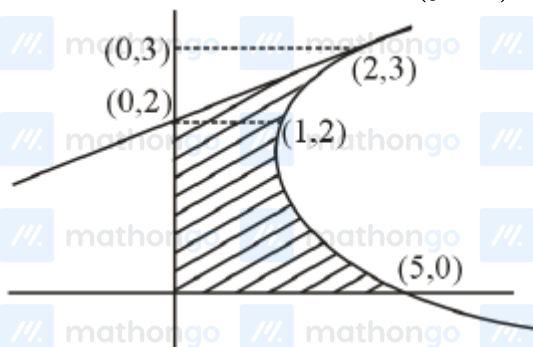
$$\Rightarrow 12Y(2) = \frac{5}{3} \times 12 = 20$$

Q10

Solving the equations

$$(y-2)^2 = x-1 \text{ and } x-2y+4=0$$

$$x = 2(y-2)$$



$$\frac{x^2}{4} = x - 1$$

$$x^2 - 4x + 4 = 0$$

$$(x-2)^2 = 0$$

$$x = 2$$

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Solutions

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Exclude area (w.r.t. y-axis) = $\int_0^3 x dy$ - Area of Δ .

$$= \int_0^3 ((y-2)^2 + 1) dy - \frac{1}{2} \times 1 \times 2$$

$$= \int_0^3 (y^2 - 4y + 5) dy - 1$$

$$= \left[\frac{y^3}{3} - 2y^2 + 5y \right]_0^3 - 1$$

$$= 9 - 18 + 15 - 1 = 5$$

Q11

$$\frac{y^2 \leq 4x, x < 4}{xy(x-1)(x-2) > 0} > 0$$

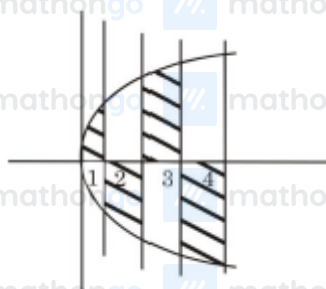
Case-I : $y > 0$

$$\frac{x(x-1)(x-2)}{(x-3)(x-4)} > 0$$

$$x \in (0, 1) \cup (2, 3)$$

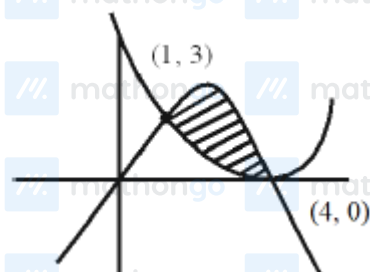
Case-II: $y < 0$

$$\frac{x(x-1)(x-2)}{(x-3)(x-4)} < 0, x \in (1, 2) \cup (3, 4)$$



$$\begin{aligned} \text{Area} &= 2 \int_0^4 \sqrt{x} dx \\ &= 2 \cdot \frac{2}{3} \left[x^{3/2} \right]_0^4 = \frac{32}{3} \end{aligned}$$

Q12



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Solutions

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$$\text{Area} = \left| \int_1^4 \left[(4x - x^2) - \frac{(x-4)^2}{3} \right] dx \right|$$

$$\text{Area} = \left| \frac{4x^2}{2} - \frac{x^3}{3} - \frac{(x-4)^3}{9} \right|_1^4$$

$$= \left| \left(\frac{64}{2} - \frac{64}{3} - \frac{4}{2} + \frac{1}{3} - \frac{27}{9} \right) \right|$$

$$\Rightarrow (27 - 21) = 6$$

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